

```
In [31]: import pandas as pd
import matplotlib.pyplot as plt

import numpy as np
```

```
In [32]: dataset =pd.read_csv("insurance.csv")
```

```
In [33]: dataset =pd.get_dummies(dataset,drop_first=True)
```

```
In [34]: dataset.columns
```

```
Out[34]: Index(['age', 'bmi', 'children', 'expenses', 'sex_male', 'smoker_yes',
               'region_northwest', 'region_southeast', 'region_southwest'],
              dtype='str')
```

```
In [35]: independent = dataset[['age', 'bmi', 'children', 'sex_male', 'smoker_yes',
                                'region_northwest', 'region_southeast', 'region_southwest']]
dependent = dataset[['expenses']]
```

```
In [36]: from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(independent,dependent,test
```

```
In [ ]: from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import r2_score
from itertools import product
import pandas as pd
```

```
# Parameter values
```

```
n_estimators_values = [50, 100, 200]
```

```
criterion_values = [
    "squared_error",
    "absolute_error",
    "friedman_mse",
    "poisson"
]
```

```
max_depth_values = [None, 5, 10, 15, 20]
```

```
min_samples_split_values = [2, 5, 10]
```

```
min_samples_leaf_values = [1, 2, 4]
```

```
max_features_values = [1.0, "sqrt", "log2"]
```

```
bootstrap_values = [True, False]
```

```
# Store all results
```

```
results = []
```

```
# Test all combinations
```

```

for (
    n_estimators,
    criterion,
    max_depth,
    min_samples_split,
    min_samples_leaf,
    max_features,
    bootstrap
) in product(
    n_estimators_values,
    criterion_values,
    max_depth_values,
    min_samples_split_values,
    min_samples_leaf_values,
    max_features_values,
    bootstrap_values
):

    regressor = RandomForestRegressor(
        n_estimators=n_estimators,
        criterion=criterion,
        max_depth=max_depth,
        min_samples_split=min_samples_split,
        min_samples_leaf=min_samples_leaf,
        max_features=max_features,
        bootstrap=bootstrap,
        random_state=42,
        n_jobs=-1
    )

    regressor.fit(x_train, y_train)

    y_pred = regressor.predict(x_test)

    r2 = r2_score(y_test, y_pred)

    # Save results

    results.append({

        "n_estimators": n_estimators,
        "criterion": criterion,
        "max_depth": max_depth,
        "min_samples_split": min_samples_split,
        "min_samples_leaf": min_samples_leaf,
        "max_features": max_features,
        "bootstrap": bootstrap,
        "R2": r2

    })

# Convert results into DataFrame

results_df = pd.DataFrame(results)

# Sort by highest R2

```

```
results_df = results_df.sort_values(
    by="R2",
    ascending=False
)

# Display ALL combinations and R2

pd.set_option("display.max_rows", None)
print(results_df.to_string(index=False))

# Save to Excel

results_df.to_excel(
    "random_forest_all_results.xlsx",
    index=False
)

# Display best combination

print("\nBEST COMBINATION:")
print(results_df.iloc[0])
```

```
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: Dat
aConversionWarning: A column-vector y was passed when a 1d array was expec
ted. Please change the shape of y to (n_samples,), for example using.ravel
().
    return fit_method(estimator, *args, **kwargs)
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: Dat
aConversionWarning: A column-vector y was passed when a 1d array was expec
ted. Please change the shape of y to (n_samples,), for example using.ravel
().
    return fit_method(estimator, *args, **kwargs)
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: Dat
aConversionWarning: A column-vector y was passed when a 1d array was expec
ted. Please change the shape of y to (n_samples,), for example using.ravel
().
    return fit_method(estimator, *args, **kwargs)
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: Dat
aConversionWarning: A column-vector y was passed when a 1d array was expec
ted. Please change the shape of y to (n_samples,), for example using.ravel
().
    return fit_method(estimator, *args, **kwargs)
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: Dat
aConversionWarning: A column-vector y was passed when a 1d array was expec
ted. Please change the shape of y to (n_samples,), for example using.ravel
().
    return fit_method(estimator, *args, **kwargs)
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: Dat
aConversionWarning: A column-vector y was passed when a 1d array was expec
ted. Please change the shape of y to (n_samples,), for example using.ravel
().
    return fit_method(estimator, *args, **kwargs)
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: Dat
aConversionWarning: A column-vector y was passed when a 1d array was expec
ted. Please change the shape of y to (n_samples,), for example using.ravel
().
    return fit_method(estimator, *args, **kwargs)
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: Dat
aConversionWarning: A column-vector y was passed when a 1d array was expec
ted. Please change the shape of y to (n_samples,), for example using.ravel
().
    return fit_method(estimator, *args, **kwargs)
```

```
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: Dat
aConversionWarning: A column-vector y was passed when a 1d array was expec
ted. Please change the shape of y to (n_samples,), for example using ravel
()).
    return fit_method(estimator, *args, **kwargs)
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: Dat
aConversionWarning: A column-vector y was passed when a 1d array was expec
ted. Please change the shape of y to (n_samples,), for example using ravel
()).
    return fit_method(estimator, *args, **kwargs)
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: Dat
aConversionWarning: A column-vector y was passed when a 1d array was expec
ted. Please change the shape of y to (n_samples,), for example using ravel
()).
    return fit_method(estimator, *args, **kwargs)
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: Dat
aConversionWarning: A column-vector y was passed when a 1d array was expec
ted. Please change the shape of y to (n_samples,), for example using ravel
()).
    return fit_method(estimator, *args, **kwargs)
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: Dat
aConversionWarning: A column-vector y was passed when a 1d array was expec
ted. Please change the shape of y to (n_samples,), for example using ravel
()).
    return fit_method(estimator, *args, **kwargs)
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: Dat
aConversionWarning: A column-vector y was passed when a 1d array was expec
ted. Please change the shape of y to (n_samples,), for example using ravel
()).
    return fit_method(estimator, *args, **kwargs)
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: Dat
aConversionWarning: A column-vector y was passed when a 1d array was expec
ted. Please change the shape of y to (n_samples,), for example using ravel
()).
    return fit_method(estimator, *args, **kwargs)
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: Dat
aConversionWarning: A column-vector y was passed when a 1d array was expec
ted. Please change the shape of y to (n_samples,), for example using ravel
()).
    return fit_method(estimator, *args, **kwargs)
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: Dat
aConversionWarning: A column-vector y was passed when a 1d array was expec
ted. Please change the shape of y to (n_samples,), for example using ravel
()).
    return fit_method(estimator, *args, **kwargs)
```



```

C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples,), for example using ravel().
    return fit_method(estimator, *args, **kwargs)
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples,), for example using ravel().
    return fit_method(estimator, *args, **kwargs)
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples,), for example using ravel().
    return fit_method(estimator, *args, **kwargs)
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples,), for example using ravel().
    return fit_method(estimator, *args, **kwargs)
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples,), for example using ravel().
    return fit_method(estimator, *args, **kwargs)
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples,), for example using ravel().
    return fit_method(estimator, *args, **kwargs)
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples,), for example using ravel().
    return fit_method(estimator, *args, **kwargs)
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.py:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.utils.parallel.Parallel` to make it possible to propagate the scikit-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.py:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.utils.parallel.Parallel` to make it possible to propagate the scikit-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.py:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.utils.parallel.Parallel` to make it possible to propagate the scikit-learn configuration of the current thread to the joblib workers.
    warnings.warn(

```

```
C:\Users\goushiik lap\anaconda3\Lib\site-packages\sklearn\u\tls\parallel.p
y:l44: UserWarning: `sklearn.u\tls.parallel.delayed` should be used with `
sklearn.u\tls.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
warnings.warn(
C:\Users\goushiik lap\anaconda3\Lib\site-packages\sklearn\u\tls\parallel.p
y:l44: UserWarning: `sklearn.u\tls.parallel.delayed` should be used with `
sklearn.u\tls.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
warnings.warn(
C:\Users\goushiik lap\anaconda3\Lib\site-packages\sklearn\u\tls\parallel.p
y:l44: UserWarning: `sklearn.u\tls.parallel.delayed` should be used with `
sklearn.u\tls.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
warnings.warn(
C:\Users\goushiik lap\anaconda3\Lib\site-packages\sklearn\u\tls\parallel.p
y:l44: UserWarning: `sklearn.u\tls.parallel.delayed` should be used with `
sklearn.u\tls.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
warnings.warn(
C:\Users\goushiik lap\anaconda3\Lib\site-packages\sklearn\u\tls\parallel.p
y:l44: UserWarning: `sklearn.u\tls.parallel.delayed` should be used with `
sklearn.u\tls.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
warnings.warn(
C:\Users\goushiik lap\anaconda3\Lib\site-packages\sklearn\u\tls\parallel.p
y:l44: UserWarning: `sklearn.u\tls.parallel.delayed` should be used with `
sklearn.u\tls.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
warnings.warn(
C:\Users\goushiik lap\anaconda3\Lib\site-packages\sklearn\u\tls\parallel.p
y:l44: UserWarning: `sklearn.u\tls.parallel.delayed` should be used with `
sklearn.u\tls.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
warnings.warn(
C:\Users\goushiik lap\anaconda3\Lib\site-packages\sklearn\u\tls\parallel.p
y:l44: UserWarning: `sklearn.u\tls.parallel.delayed` should be used with `
sklearn.u\tls.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
warnings.warn(
C:\Users\goushiik lap\anaconda3\Lib\site-packages\sklearn\u\tls\parallel.p
y:l44: UserWarning: `sklearn.u\tls.parallel.delayed` should be used with `
sklearn.u\tls.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
warnings.warn(
```

```
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: Dat
aConversionWarning: A column-vector y was passed when a 1d array was expec
ted. Please change the shape of y to (n_samples,), for example using ravel
().
    return fit_method(estimator, *args, **kwargs)
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
```

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

```
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: Dat
aConversionWarning: A column-vector y was passed when a 1d array was expec
ted. Please change the shape of y to (n_samples,), for example using ravel
().
    return fit_method(estimator, *args, **kwargs)
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
```

[illegible]

[illegible]

[illegible]

```
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: Dat
aConversionWarning: A column-vector y was passed when a 1d array was expec
ted. Please change the shape of y to (n_samples,), for example using ravel
().
    return fit_method(estimator, *args, **kwargs)
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: Dat
aConversionWarning: A column-vector y was passed when a 1d array was expec
ted. Please change the shape of y to (n_samples,), for example using ravel
().
    return fit_method(estimator, *args, **kwargs)
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
```

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

```
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p  
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `  
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik  
it-learn configuration of the current thread to the joblib workers.  
    warnings.warn(  
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: Dat  
aConversionWarning: A column-vector y was passed when a 1d array was expec  
ted. Please change the shape of y to (n_samples,), for example using ravel  
( ).  
    return fit_method(estimator, *args, **kwargs)  
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p  
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `  
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik  
it-learn configuration of the current thread to the joblib workers.  
    warnings.warn(  
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p  
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `  
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik  
it-learn configuration of the current thread to the joblib workers.  
    warnings.warn(  
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p  
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `  
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik  
it-learn configuration of the current thread to the joblib workers.  
    warnings.warn(  
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p  
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `  
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik  
it-learn configuration of the current thread to the joblib workers.  
    warnings.warn(  
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p  
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `  
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik  
it-learn configuration of the current thread to the joblib workers.  
    warnings.warn(  
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p  
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `  
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik  
it-learn configuration of the current thread to the joblib workers.  
    warnings.warn(  
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p  
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `  
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik  
it-learn configuration of the current thread to the joblib workers.  
    warnings.warn(  
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p  
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `  
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik  
it-learn configuration of the current thread to the joblib workers.
```

[illegible]

[illegible]

[illegible]

[illegible]

```
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
```

```
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
```

[illegible]

```

C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: Dat
aConversionWarning: A column-vector y was passed when a 1d array was expec
ted. Please change the shape of y to (n_samples,), for example using ravel
().
    return fit_method(estimator, *args, **kwargs)
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(

```

[illegible]

[illegible]

```
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: Dat
aConversionWarning: A column-vector y was passed when a 1d array was expec
ted. Please change the shape of y to (n_samples,), for example using ravel
().
    return fit_method(estimator, *args, **kwargs)
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: Dat
aConversionWarning: A column-vector y was passed when a 1d array was expec
ted. Please change the shape of y to (n_samples,), for example using ravel
().
    return fit_method(estimator, *args, **kwargs)
```

```
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples,), for example using ravel().
```

```
    return fit_method(estimator, *args, **kwargs)
```

```
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples,), for example using ravel().
```

```
    return fit_method(estimator, *args, **kwargs)
```

```
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.py:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.utils.parallel.Parallel` to make it possible to propagate the scikit-learn configuration of the current thread to the joblib workers.
```

```
    warnings.warn(
```

```
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.py:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.utils.parallel.Parallel` to make it possible to propagate the scikit-learn configuration of the current thread to the joblib workers.
```

```
    warnings.warn(
```

```
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.py:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.utils.parallel.Parallel` to make it possible to propagate the scikit-learn configuration of the current thread to the joblib workers.
```

```
    warnings.warn(
```

```
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.py:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.utils.parallel.Parallel` to make it possible to propagate the scikit-learn configuration of the current thread to the joblib workers.
```

```
    warnings.warn(
```

```
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.py:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.utils.parallel.Parallel` to make it possible to propagate the scikit-learn configuration of the current thread to the joblib workers.
```

```
    warnings.warn(
```

```
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.py:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.utils.parallel.Parallel` to make it possible to propagate the scikit-learn configuration of the current thread to the joblib workers.
```

```
    warnings.warn(
```

```
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.py:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.utils.parallel.Parallel` to make it possible to propagate the scikit-learn configuration of the current thread to the joblib workers.
```

```
    warnings.warn(
```

```
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.py:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.utils.parallel.Parallel` to make it possible to propagate the scikit-learn configuration of the current thread to the joblib workers.
```

```
    warnings.warn(
```

```
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.py:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.utils.parallel.Parallel` to make it possible to propagate the scikit-learn configuration of the current thread to the joblib workers.
```

```
    warnings.warn(
```

```
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.py:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.utils.parallel.Parallel` to make it possible to propagate the scikit-learn configuration of the current thread to the joblib workers.
```

```
    warnings.warn(
```

[illegible]

```
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
```

[illegible]

[illegible]

```
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: Dat
aConversionWarning: A column-vector y was passed when a 1d array was expec
ted. Please change the shape of y to (n_samples,), for example using ravel
().
    return fit_method(estimator, *args, **kwargs)
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
```

[illegible]

```
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
```

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

```
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: Dat
aConversionWarning: A column-vector y was passed when a 1d array was expec
ted. Please change the shape of y to (n_samples,), for example using ravel
().
    return fit_method(estimator, *args, **kwargs)
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
```

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

```
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples,), for example using ravel().
```

```
    return fit_method(estimator, *args, **kwargs)
```

```
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.py:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.utils.parallel.Parallel` to make it possible to propagate the scikit-learn configuration of the current thread to the joblib workers.
```

```
    warnings.warn(
```

```
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.py:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.utils.parallel.Parallel` to make it possible to propagate the scikit-learn configuration of the current thread to the joblib workers.
```

```
    warnings.warn(
```

```
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.py:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.utils.parallel.Parallel` to make it possible to propagate the scikit-learn configuration of the current thread to the joblib workers.
```

```
    warnings.warn(
```

```
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.py:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.utils.parallel.Parallel` to make it possible to propagate the scikit-learn configuration of the current thread to the joblib workers.
```

```
    warnings.warn(
```

```
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.py:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.utils.parallel.Parallel` to make it possible to propagate the scikit-learn configuration of the current thread to the joblib workers.
```

```
    warnings.warn(
```

```
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.py:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.utils.parallel.Parallel` to make it possible to propagate the scikit-learn configuration of the current thread to the joblib workers.
```

```
    warnings.warn(
```

```
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.py:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.utils.parallel.Parallel` to make it possible to propagate the scikit-learn configuration of the current thread to the joblib workers.
```

```
    warnings.warn(
```

```
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.py:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.utils.parallel.Parallel` to make it possible to propagate the scikit-learn configuration of the current thread to the joblib workers.
```

```
    warnings.warn(
```

```
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.py:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.utils.parallel.Parallel` to make it possible to propagate the scikit-learn configuration of the current thread to the joblib workers.
```

```
    warnings.warn(
```

```
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.py:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.utils.parallel.Parallel` to make it possible to propagate the scikit-learn configuration of the current thread to the joblib workers.
```

```
    warnings.warn(
```

```
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.py:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `sklearn.utils.parallel.Parallel` to make it possible to propagate the scikit-learn configuration of the current thread to the joblib workers.
```

```
    warnings.warn(
```

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

```
C:\Users\goushi lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushi lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushi lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushi lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushi lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushi lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushi lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushi lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushi lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushi lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: Dat
aConversionWarning: A column-vector y was passed when a 1d array was expec
ted. Please change the shape of y to (n_samples,), for example using ravel
()).
```

[illegible]

[illegible]

[illegible]

[illegible]

```
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\parallel.p
y:144: UserWarning: `sklearn.utils.parallel.delayed` should be used with `
sklearn.utils.parallel.Parallel` to make it possible to propagate the scik
it-learn configuration of the current thread to the joblib workers.
    warnings.warn(
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: Dat
aConversionWarning: A column-vector y was passed when a 1d array was expec
ted. Please change the shape of y to (n_samples,), for example using.ravel
().
    return fit_method(estimator, *args, **kwargs)
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: Dat
aConversionWarning: A column-vector y was passed when a 1d array was expec
ted. Please change the shape of y to (n_samples,), for example using.ravel
().
    return fit_method(estimator, *args, **kwargs)
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: Dat
aConversionWarning: A column-vector y was passed when a 1d array was expec
ted. Please change the shape of y to (n_samples,), for example using.ravel
().
    return fit_method(estimator, *args, **kwargs)
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: Dat
aConversionWarning: A column-vector y was passed when a 1d array was expec
ted. Please change the shape of y to (n_samples,), for example using.ravel
().
    return fit_method(estimator, *args, **kwargs)
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: Dat
aConversionWarning: A column-vector y was passed when a 1d array was expec
ted. Please change the shape of y to (n_samples,), for example using.ravel
().
    return fit_method(estimator, *args, **kwargs)
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: Dat
aConversionWarning: A column-vector y was passed when a 1d array was expec
ted. Please change the shape of y to (n_samples,), for example using.ravel
().
    return fit_method(estimator, *args, **kwargs)
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: Dat
aConversionWarning: A column-vector y was passed when a 1d array was expec
ted. Please change the shape of y to (n_samples,), for example using.ravel
().
    return fit_method(estimator, *args, **kwargs)
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: Dat
aConversionWarning: A column-vector y was passed when a 1d array was expec
ted. Please change the shape of y to (n_samples,), for example using.ravel
().
    return fit_method(estimator, *args, **kwargs)
```

```
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: Dat
aConversionWarning: A column-vector y was passed when a 1d array was expec
ted. Please change the shape of y to (n_samples,), for example using.ravel
().
    return fit_method(estimator, *args, **kwargs)
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: Dat
aConversionWarning: A column-vector y was passed when a 1d array was expec
ted. Please change the shape of y to (n_samples,), for example using.ravel
().
    return fit_method(estimator, *args, **kwargs)
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: Dat
aConversionWarning: A column-vector y was passed when a 1d array was expec
ted. Please change the shape of y to (n_samples,), for example using.ravel
().
    return fit_method(estimator, *args, **kwargs)
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: Dat
aConversionWarning: A column-vector y was passed when a 1d array was expec
ted. Please change the shape of y to (n_samples,), for example using.ravel
().
    return fit_method(estimator, *args, **kwargs)
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: Dat
aConversionWarning: A column-vector y was passed when a 1d array was expec
ted. Please change the shape of y to (n_samples,), for example using.ravel
().
    return fit_method(estimator, *args, **kwargs)
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: Dat
aConversionWarning: A column-vector y was passed when a 1d array was expec
ted. Please change the shape of y to (n_samples,), for example using.ravel
().
    return fit_method(estimator, *args, **kwargs)
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: Dat
aConversionWarning: A column-vector y was passed when a 1d array was expec
ted. Please change the shape of y to (n_samples,), for example using.ravel
().
    return fit_method(estimator, *args, **kwargs)
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: Dat
aConversionWarning: A column-vector y was passed when a 1d array was expec
ted. Please change the shape of y to (n_samples,), for example using.ravel
().
    return fit_method(estimator, *args, **kwargs)
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: Dat
aConversionWarning: A column-vector y was passed when a 1d array was expec
ted. Please change the shape of y to (n_samples,), for example using.ravel
().
    return fit_method(estimator, *args, **kwargs)
```



```
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: Dat
aConversionWarning: A column-vector y was passed when a 1d array was expec
ted. Please change the shape of y to (n_samples,), for example using.ravel
().
    return fit_method(estimator, *args, **kwargs)
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: Dat
aConversionWarning: A column-vector y was passed when a 1d array was expec
ted. Please change the shape of y to (n_samples,), for example using.ravel
().
    return fit_method(estimator, *args, **kwargs)
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: Dat
aConversionWarning: A column-vector y was passed when a 1d array was expec
ted. Please change the shape of y to (n_samples,), for example using.ravel
().
    return fit_method(estimator, *args, **kwargs)
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: Dat
aConversionWarning: A column-vector y was passed when a 1d array was expec
ted. Please change the shape of y to (n_samples,), for example using.ravel
().
    return fit_method(estimator, *args, **kwargs)
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: Dat
aConversionWarning: A column-vector y was passed when a 1d array was expec
ted. Please change the shape of y to (n_samples,), for example using.ravel
().
    return fit_method(estimator, *args, **kwargs)
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: Dat
aConversionWarning: A column-vector y was passed when a 1d array was expec
ted. Please change the shape of y to (n_samples,), for example using.ravel
().
    return fit_method(estimator, *args, **kwargs)
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: Dat
aConversionWarning: A column-vector y was passed when a 1d array was expec
ted. Please change the shape of y to (n_samples,), for example using.ravel
().
    return fit_method(estimator, *args, **kwargs)
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: Dat
aConversionWarning: A column-vector y was passed when a 1d array was expec
ted. Please change the shape of y to (n_samples,), for example using.ravel
().
    return fit_method(estimator, *args, **kwargs)
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: Dat
aConversionWarning: A column-vector y was passed when a 1d array was expec
ted. Please change the shape of y to (n_samples,), for example using.ravel
().
    return fit_method(estimator, *args, **kwargs)
```

```
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: Dat
aConversionWarning: A column-vector y was passed when a 1d array was expec
ted. Please change the shape of y to (n_samples,), for example using ravel
()).
    return fit_method(estimator, *args, **kwargs)
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: Dat
aConversionWarning: A column-vector y was passed when a 1d array was expec
ted. Please change the shape of y to (n_samples,), for example using ravel
()).
    return fit_method(estimator, *args, **kwargs)
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: Dat
aConversionWarning: A column-vector y was passed when a 1d array was expec
ted. Please change the shape of y to (n_samples,), for example using ravel
()).
    return fit_method(estimator, *args, **kwargs)
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: Dat
aConversionWarning: A column-vector y was passed when a 1d array was expec
ted. Please change the shape of y to (n_samples,), for example using ravel
()).
    return fit_method(estimator, *args, **kwargs)
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: Dat
aConversionWarning: A column-vector y was passed when a 1d array was expec
ted. Please change the shape of y to (n_samples,), for example using ravel
()).
    return fit_method(estimator, *args, **kwargs)
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: Dat
aConversionWarning: A column-vector y was passed when a 1d array was expec
ted. Please change the shape of y to (n_samples,), for example using ravel
()).
    return fit_method(estimator, *args, **kwargs)
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: Dat
aConversionWarning: A column-vector y was passed when a 1d array was expec
ted. Please change the shape of y to (n_samples,), for example using ravel
()).
    return fit_method(estimator, *args, **kwargs)
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: Dat
aConversionWarning: A column-vector y was passed when a 1d array was expec
ted. Please change the shape of y to (n_samples,), for example using ravel
()).
    return fit_method(estimator, *args, **kwargs)
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: Dat
aConversionWarning: A column-vector y was passed when a 1d array was expec
ted. Please change the shape of y to (n_samples,), for example using ravel
()).
    return fit_method(estimator, *args, **kwargs)
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: Dat
aConversionWarning: A column-vector y was passed when a 1d array was expec
ted. Please change the shape of y to (n_samples,), for example using ravel
()).
    return fit_method(estimator, *args, **kwargs)
```



```
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples,), for example using ravel().
    return fit_method(estimator, *args, **kwargs)
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples,), for example using ravel().
    return fit_method(estimator, *args, **kwargs)
C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\base.py:1403: DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples,), for example using ravel().
    return fit_method(estimator, *args, **kwargs)
```

```
In [29]: y_pred=regressor.predict(x_test)
         y_pred
```

```
Out[29]: array([ 6923.09978,  5191.37287, 25340.87372, 13951.84763,  3900.36877,
 6830.66395, 10642.00963, 20194.42244, 12076.40556, 20784.69107,
 6878.41844,  3311.96492,  7228.11052,  7948.94376,  9895.23247,
17659.71144,  4230.6432 , 17333.45054, 11830.5125 , 20374.24327,
40488.63884,  2454.92045,  2596.66697,  6155.71438,  7217.92014,
14112.18988, 12288.39725, 13197.9891 ,  4734.09649, 22382.064 ,
 1264.33737, 40192.79649,  8112.70324, 13669.05469, 11581.36733,
10748.59583, 13491.72607, 11418.7172 ,  9741.25092, 16441.41724,
14375.99527, 11887.17015,  5265.87824,  8193.50272,  3066.23173,
 7078.18637,  8503.18173, 40160.66782, 12947.66718, 11875.13429,
42300.99531,  4241.43764,  7657.38148, 14414.25172, 11001.76546,
13512.14093, 17055.74879,  6826.21412,  6845.93945, 41268.91312,
 1708.61701, 24008.68034, 28458.21236,  2960.30318, 18290.08897,
 8673.48727,  9245.84895,  7227.75581, 11007.10009,  8745.61768,
 7218.33316,  2567.71778,  7053.08013,  7013.54906, 10202.54143,
11212.47177,  5070.95617,  7611.29405,  1631.35976, 10845.66344,
10912.24922,  9440.66012, 18146.42415, 12424.96819, 23762.09697,
 4045.83386,  2765.0455 ,  6501.73467,  6064.57201, 10030.44711,
35410.47791, 39926.33414,  9162.03689, 46493.04061, 13760.01592,
 4720.05888,  6884.72899,  9969.5798 ,  6362.22471, 13793.0596 ,
 6886.06366,  2477.70505,  7476.57485,  6825.69383,  9864.48693,
11418.63966, 20943.5141 , 35317.76741, 23923.60058,  6620.26304,
11957.23755,  7715.67979,  4993.33971, 18645.61992, 38769.60717,
21140.8893 ,  4611.4572 , 41345.39713, 10246.0419 ,  4293.65198,
40113.79432,  9716.08458,  7655.33078,  1763.54534, 47227.17919,
 5026.28484, 12176.09942,  9137.04085, 13459.23667, 15991.30212,
10348.08797, 27684.07812, 22110.0145 , 28489.10613, 10453.23318,
19905.17114, 13416.1767 , 11936.09414, 36884.91436, 10983.9548 ,
 6539.74112, 20793.37394, 15690.78933,  3908.6755 ,  4894.13305,
37016.33819, 16043.67768,  9454.16089,  9115.79333, 13028.39895,
 5225.54546, 10729.7114 ,  1682.78695, 46609.04493, 17091.79945,
 1600.63467, 13303.23553,  1820.69603, 18846.45535, 28534.33027,
13125.20047,  6126.94034,  8868.38573, 16362.74866,  3965.48645,
49075.42343,  7566.98097,  7979.87732,  3039.10922, 11328.22642,
10499.30563, 10017.7553 ,  5739.65662,  5577.81422, 19448.09815,
 6801.60962,  8653.41108, 12226.12425,  7646.95704, 16698.23604,
47807.58527, 11883.0301 ,  4778.04156,  3122.67317, 35348.10537,
 6150.19751, 28467.01096, 11300.13189, 20992.37642,  8701.26886,
10670.465 , 12424.77864,  5183.14565,  7490.84834, 12178.08944,
10407.53106, 17780.95418,  5443.00144, 16464.99884,  6669.1673 ,
 5091.88597,  1708.7253 , 14228.69316,  9548.127 ,  1669.60833,
47057.99087, 18276.90678, 16225.36607, 44420.57124, 2355.83059,
 9351.4368 , 41371.94651,  6499.45132,  6781.94685, 38432.95275,
 6796.53496,  7147.12488,  3915.19841, 16758.35067, 11639.2842 ,
 5717.04782,  5687.47396,  2583.76887, 16181.4406 ,  5578.1618 ,
16466.501 , 14956.75233,  2042.92568, 14119.81386, 35878.68638,
 6215.39834, 18531.47821, 10305.6121 , 37699.21125, 12812.43819,
 4413.53914, 13556.01301, 28639.62615,  2214.96507, 12259.88461,
11129.01826, 24804.93874, 10101.82139, 39119.06424, 13772.94352,
 3139.71909, 13686.02499,  7025.00992, 14829.75671,  2590.77476,
 5013.48568, 17289.14288, 45252.31886,  5457.43773, 11714.48703,
15300.14405, 12033.47397,  6612.62131,  8668.20939,  9662.95544,
 6420.9479 , 24151.89221,  6708.92989, 41757.77626,  5174.0486 ,
10580.15986, 15385.75561,  8602.6972 ])
```

```
In [30]: from sklearn.metrics import r2_score
percentage=r2_score(y_test,y_pred)*100
print(f"result: {percentage} %")
```

```
result: 80.52300354276625 %
```

